Alex Olivia Koehn [**Rutgers University** - Bachelors of Science - Computer Science]

Software Developer - Systems · Backend · Network Infrastructure - 📞 (609)-922-7835

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Skills

10+ years personal programming experience
Java, Kotlin, Python
C, HTML/CSS, Javascript
SQL/SQLite, Assembly, Bash,
AI, ML, Object Oriented Design Principals

Tools and Platforms

Git, Docker, Linux, Nginx, Fail2Ban, Crowdsec, Cloudflare, JavaFX, Jetpack Compose, Unraid, Microsoft 365, Google Workspace, Maven

Concepts

REST APIs, Virtualization, OOP, Networking, REverse Proxying, SSL/TLS, Data backup and parity, Self-hosted infrastructure, SMB/network shares, Intrusion detection

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Home Lab - Self Hosted Linux Server Unraid - Docker - Linux

- Hands-on experience deploying and maintaining 10+ Dockerised services including Plex, Tunarr, Nginx Proxy Manager, and Handbrake on an Unraid NAS
- Configured Nginx reverse proxy with SSL termination via Cloudflare, exposing services securely over HTTPS
- Hardened server security with Fail2ban and CrowdSec for automated intrusion detection and IP banning
- Provisioned and manage VMs for cross-platform binary compilation
- Self-hosted internet facing services with 10+ users with custom configurations, dependency management, memory tuning, and permissions controls
- Configured Unraid file shares with SMB, mapping persistent network drives on both Windows (mapped drives) and Linux clients
- Managed shared smb resources, users, groups, and file permissions across shared environments; maintained parity drive redundancy for data protection
- Bash scripts to automate routine maintenance and administration tasks

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NASA Solar Event Tracker Java · JavaFX · REST APIs

- Built a frontend desktop application that fetches and displays live solar event data from NASA's public REST API
- Parsed JSON API responses into typed Java classes using object-oriented design patterns
- Designed a JavaFX UI to present structured event data in a clear, user-friendly format
- Incorporates third-party dependencies using Maven

General Experience

- Developed web scraper in C that dynamically traverses web pages and crawls additional links as discovered. Stored discovered links in a hash table for efficiency and to prevent recrawling. Developed as a final project for my C programming language course.

- Developed Solar Flare dashboard in Java with Java FX as the UI fronted with dependency management using Maven. Leveraged the official NASA REST api to request official solar flare satellite data, deserialized into Java structures and presented to the user in a clean UI with exceptions handled. Incorporates third party dependencies Jackson Core and Jackson databind for efficiency. Developed as a final project for my JAVA and API course.